

ARPIT KUMAR

ML Engineer & AI Researcher | IIT Kharagpur | NLP · LLM/RAG · MLOps · Production ML

+91-9506035861 | arpitkumar@kgpian.iitkgp.ac.in | arpitkumar.dev | arpitkumar2004 | arpit-kumar-shivam | Lucknow, India

PROFESSIONAL SUMMARY

Final-year Dual Degree student at IIT Kharagpur (Chem Engg + AI and HPC, GPA 8.86) graduating April 2027, with end-to-end ML experience across LLM and RAG engineering, NLP, and production MLOps. Engineered a Tri-Path Multimodal RAG framework, improving retrieval baselines by 13.7–19.8% across 10 set benchmarks and mitigating 91% of known RAG failure modes. Deployed an LLM support platform, cutting resolution time by 35% and reduced hallucination rate by 40%. Ranked Top 0.5% globally in Amazon ML Challenge 2025 (50K+ participant) with reproducible multimodal prediction system, achieved 93.09% test accuracy on Speech AI NLI research, outperform ResNet baselines. Codeforces Expert (peak 1612), AIR 135 in Integral Cup 2026 S1, and Gold Medalist in IIT KGP's GC Data Analytics 2025. Seeking a full-time ML Engineering role to build and ship production AI systems with measurable real-world impact

EDUCATION

Indian Institute of Technology (IIT) Kharagpur | Kharagpur, India Jul 2022 – Apr 2027
Integrated B.Tech – M.Tech (Dual Degree) in Chemical Engineering GPA : 8.86/10
Specialization: Artificial Intelligence (AI) and High Performance Computing (HPC)

WORK EXPERIENCE

AI Researcher | Internship May 2026 – Jul 2026
Sponsored Research & Industrial Consultancy (SRIC), IIT Kharagpur On-site • Kharagpur, West Bengal, IN
Built a Tri-Path Multimodal RAG framework for documents by combining dense BM25S retrieval, DuckDB Text-to-SQL, BLIP-2 vision retrieval, weighted RRF, cross-encoder reranking, and NLI attribution, by production-oriented Python architecture with scalable indexing and modular ingestion components
Improved retrieval and reasoning performance by integrating soft probabilistic routing with parallel text, table, and vision path, achieved 0.612 Recall@5, 0.552 nDCG@5, 0.578 SQL execution success, and 0.657 attribution precision, improving RAG baselines by 13.7%–19.8% points across 10 benchmarks
Productionized the framework through PyPI packaging, GitHub Actions CI, automated vulnerability auditing, multi-version testing, and a 22-dimensional evaluation harness, validated the architecture across 10 benchmark datasets and mitigated 91% of known RAG failure modes during robustness testing
Technologies / Skills Used : PyTorch, Transformers, FastAPI, Qdrant, DuckDB, BM25s, Docling, PyPI

AI Researcher (UG) | Internship May 2025 – Jul 2025
Advanced Technology Development Centre | Prof. Shyamal Kumar Das Mandal | IIT Kharagpur On-site • Kharagpur, West Bengal, IN
Developed two-phase native language identification framework using fixed Vision Transformer on L2 English speech, integrating Mel-spectrograms, F0 contours, and fluency descriptors to isolate acoustic and temporal cues across Hindi, Kannada, Malayalam, Tamil, and Telugu speakers in NISP datasets
Engineered 128-Mel spectrograms, log-F0, and 14-feature fluency representations from 44.1 kHz speech, trained ViT-small models with AdamW and cosine scheduling, implemented PyTorch feature caching to reduce epoch time from approximately 13 minute to 30 seconds for efficient model training
Achieved 93.09% test accuracy with Mel-spectrograms, outperforming CNN and ResNet-18 baselines by 6.00% and 4.80%, stratified fluency analysis showed accuracy declining from 88.01% at low fluency to 84.44% at high fluency, revealing weaker accent cues in highly fluent speech overall in dataset
Technologies / Skills Used : CNN (ResNet), BiLSTM, Mel-Spectrograms, MFCCs, t-SNE, Wav2Vec2, WavLM

PROJECTS

PrismPrice — Multimodal Product Price Intelligence Platform | Github | Live Demo | Website Sep 2025 – Dec 2025
AI Engineer | Top 0.5%, Amazon ML Challenge'25 — national competition (sponsored by Amazon)
Built a multimodal product-price prediction pipeline across 3 feature modalities—TF-IDF/SBERT text, CLIP/ResNet vision, and unit signals, feeding 5+ model families with cross-validation and OOF stacking, plus persisted feature transformations, to create an end-to-end, production-ready pricing system
Ensured reproducible offline-to-online inference by packaging model weights, transformers, schema rules, and preprocessing into immutable run-scoped bundles, implemented versioned registry with staging/production promotion, deployment manifests, audit trails, and rollback support for governed model
Productionize real-time predictions through FastAPI endpoints, latency and error monitoring, and GitHub Actions automation, with integrated MLflow and DVC tracking, scheduled retraining, deployment validation, Hugging Face Spaces delivery with automated live health checks and model promotion gates
Technologies / Tools Used : PyTorch, BERT, CLIP, Docker, Python, FastAPI, MLflow

LLM-Powered Employee Support Platform with RAG and Scalable AI Infrastructure | Github Jan 2025 – Apr 2025
AI Engineer | 1st place, GC OpenSoft'25 — inter-hall software competition, IIT Kharagpur (sponsored by Deloitte)
Reduced employee-support resolution time by 35% by building an LLM-powered RAG platform with GPT-4o, LangChain, LangGraph, vector retrieval, and 5 specialized agents, enabling personalized employee analysis, context-aware recommendations, and faster assistance across diverse workplace support
Reduced hallucination rate by 40% by designing a dual-knowledge RAG counseling pipeline combining employee records, psychological question banks, session-aware context, sentiment-driven branching, structured citation-based generation for reliable multi-turn conversations and automatic escalation
Boosted API throughput by 50% by deploying async FastAPI services with concurrent agent execution, non-blocking LLM communication, optimizing retrieval workflows, and production infrastructure using Docker, Kubernetes, and GCP, with autoscaling, load balancing, CI/CD, and real-time monitoring
Technologies / Tools Used : FastAPI, Python, LangChain, OpenAI API, Groq, PyTorch, Docker, JWT

SKILLS

Programming Languages :	Python, C++, SQL, C, JavaScript
Frameworks & Libraries :	PyTorch, TensorFlow, Scikit-learn, FastAPI, Flask, JAX, Numpy , Pandas, Matplotlib, Seaborn, React
Data Science :	Machine Learning, Statistics, Probability, A/B Testing, Deep learning, Data Analysis, Data Visualization
Generative AI :	Retrieval-Augmented Generation, Model Interpretability, Distributed Inference, Quantization
Tools & Platforms :	MLflow, DVC, Git, GitHub, Docker, Dagshub, CI/CD, GitHub Actions
Databases :	MySQL, PostgreSQL

POSITION OF RESPONSIBILITY

Developers' Society, TSG, IIT Kharagpur Sep 2023 – Present
Advisor | Machine Learning & Software Architecture Kharagpur, West Bengal, IN

AWARDS & ACHIEVEMENTS

- Competitive Programming:** Achieved Expert rating on Codeforces (Peak: 1612), advance proficiency in data structures and algorithmic problem-solving
- Quant & ML Challenges:** Ranked Top 20 national in DTL Quant Challenge 2024 & Top 0.5% global Amazon ML Challenge 2025 out of 50K+ participant
- Mathematics:** Secured AIR 135 in the Integral Cup 2026 S1 in 3 tracks Probability Theory & Statistics, Linear Algebra & Optimization and Integral Analysis
- Institute Championships:** Awarded 1 Gold Medal (General Championship Data Analytics 2025), 2 Silver Medals (Open IIT Data Analytics 2024 & Open IIT Case Study 2024), and 1 Bronze Medal (General Championship ChemQuest 2025) at IIT Kharagpur